



Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

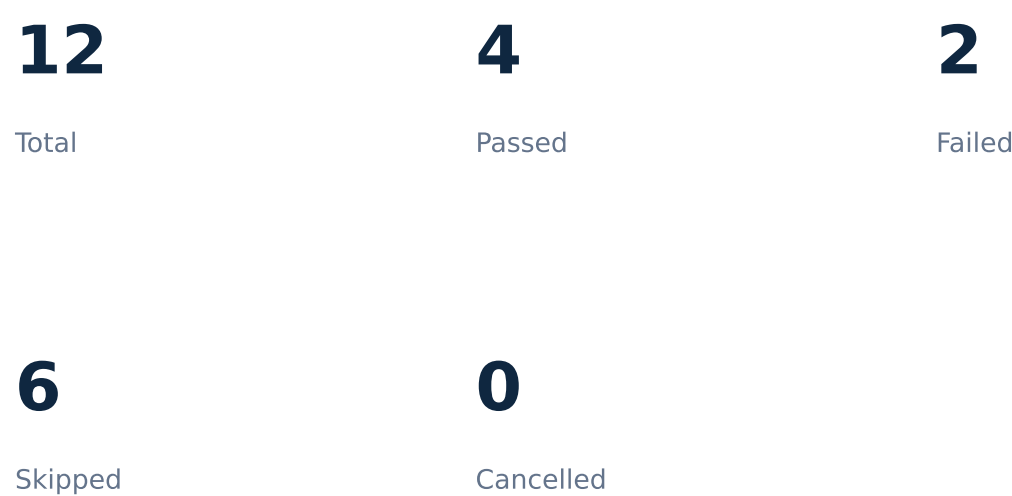
20260813_121702_d61518

Date

2026-08-13T11:17:02.525784+00:00

Executive Summary

20260813_121702_d61518



Overall measured availability: 72.45%

During this campaign, 4 of 12 scenarios passed, 2 failed and 6 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

Test Environment

Architecture and technologies

Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

Test Scenarios

Scenario	kubernetes	swarm
container-kill	1	1
cpu	1	1
latency	1	1
memory	1	1
network-partition	1	1
node-failure	1	1

Full Results Table

Scenario	Platform	Detection	Recovery	Required	Success	Avg ms	P95 ms	HTTP Fail	Avail. %	Status	Skip Reason	Error Message
container-kill	swarm	1.765	11.359	True	True	10.198	21.221	9	25.0	PASS	N/A	N/A
node-failure	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Single-node Docker Swarm cannot demonstrate failover routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/EST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured	N/A
cpu	swarm	N/A	N/A	N/A	Not Required	9.224	21.032	N/A	100.0	PASS	N/A	N/A
memory	swarm	N/A	N/A	N/A	N/A	9.821	23.463	N/A	100.0	FAIL	Stress pod finished with phase Running.	container exceeded its scenario
network-partition	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Single-node Docker Swarm cannot demonstrate failover routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/EST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured	N/A
latency	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Single-node Docker Swarm cannot demonstrate failover routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/EST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured	N/A
container-kill	kubernetes	6.953	19.094	True	True	1365.522	2050.237	7	22.22	PASS	N/A	N/A
node-failure	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Multi-node kubelet does not provide a safe backend to store failover routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/EST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured	N/A
cpu	kubernetes	N/A	N/A	N/A	Not Required	21.771	38.8	N/A	100.0	PASS	N/A	N/A
memory	kubernetes	N/A	N/A	N/A	N/A	56131.498	448873.396	1	87.5	FAIL	Stress pod finished with phase Running.	N/A
network-partition	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Multi-node kubelet does not provide a safe backend to store failover routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/EST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured	N/A
latency	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Multi-node kubelet does not provide a safe backend to store failover routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/EST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured	N/A

swarm — container-kill

State: **Pass**
Detection Time (s): 1.765
Recovery Time (s): 11.359

HTTP Success: N/A
HTTP Failures: 9
Availability (%): 25.0
CPU (%): N/A
Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — node-failure

State: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Single-node Docker Swarm cannot demonstrate real node rescheduling.

swarm — cpu

Status: Pass
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — memory

State: Fail
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Swarm stress container exceeded its scenario deadline.

swarm — network-partition

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

swarm — latency

State: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

kubernetes — container-kill

State: Pass
Detection Time (s): 6.953

Recovery Time (s): 19.094

HTTP Success: N/A

HTTP Failures: 7

Availability (%): 22.22

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — node-failure

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL path after rescheduling.

kubernetes — cpu

Status: Pass
Detection Time (s): N/A
Recovery Time (s): N/A
HTTP Success: N/A
HTTP Failures: N/A
Availability (%): 100.0
CPU (%): N/A
Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — memory

State: **Fail**
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: 1

Availability (%): 87.5

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Stress pod finished with phase Running.

kubernetes — network-partition

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

kubernetes — latency

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

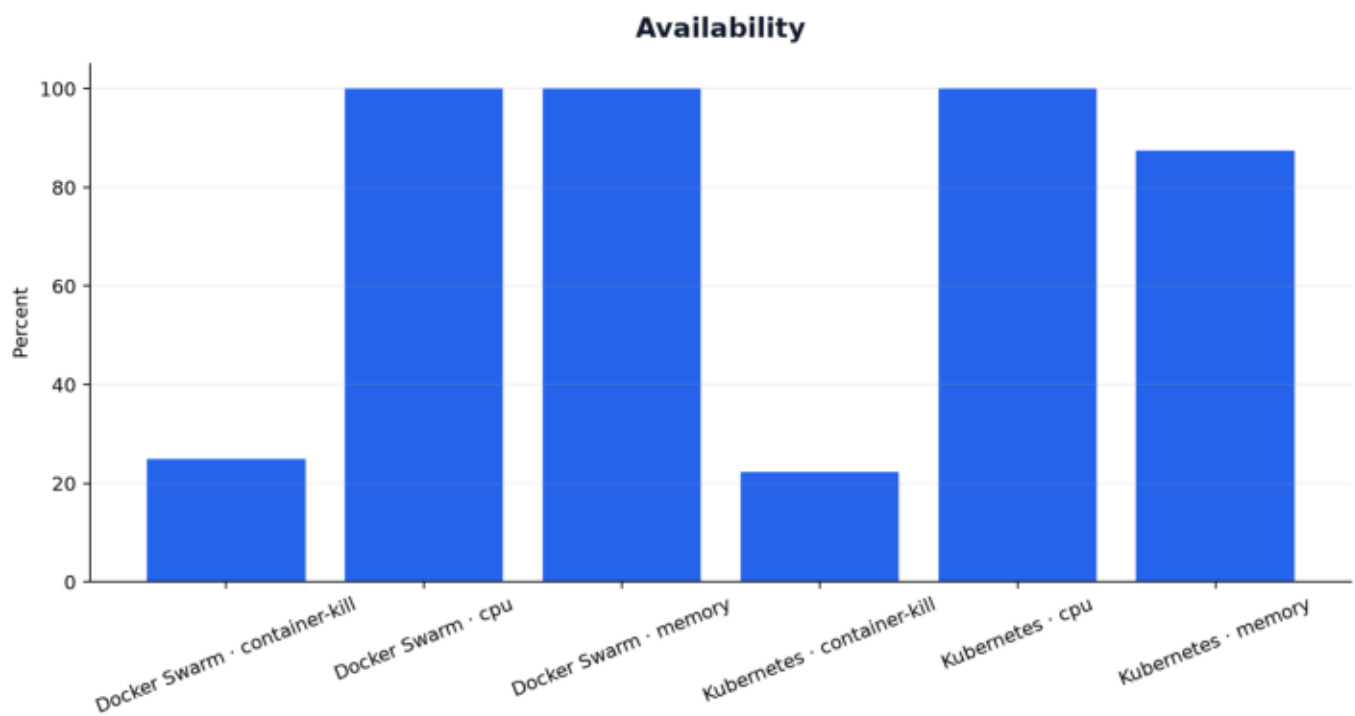
Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

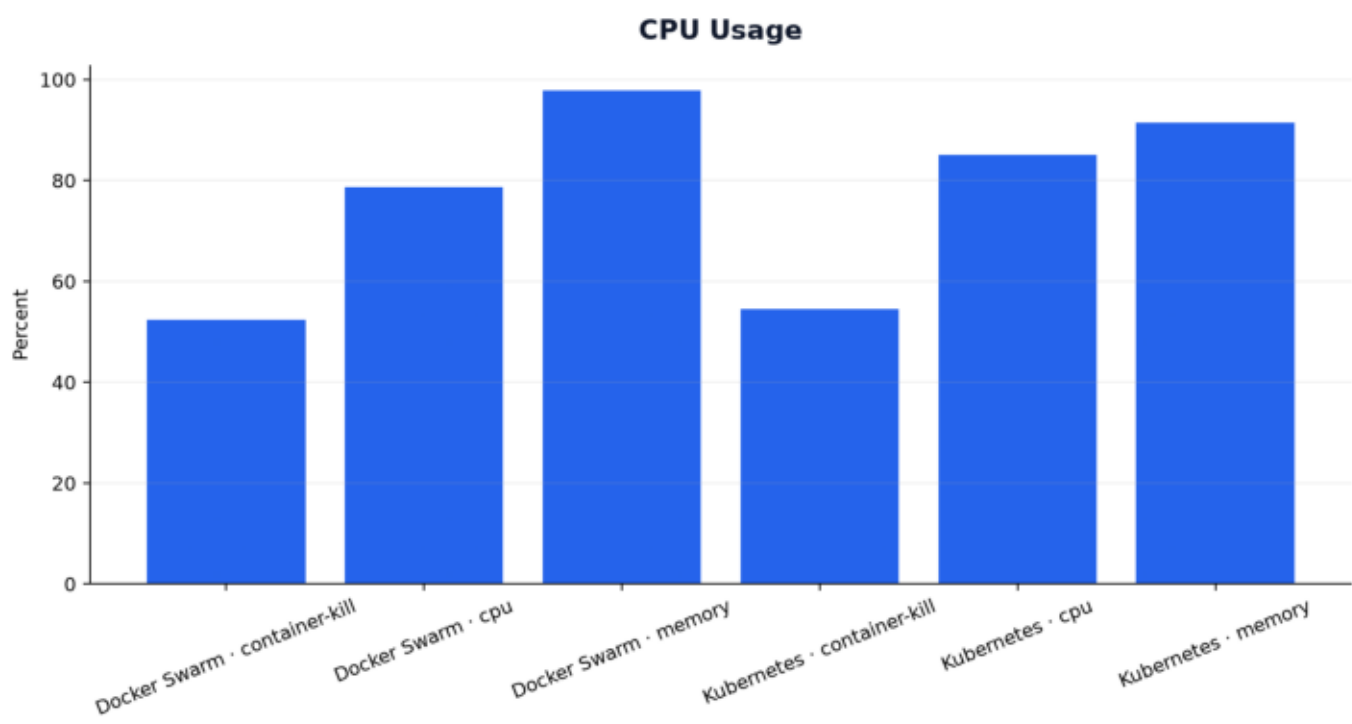
Availability

Current-session measured data



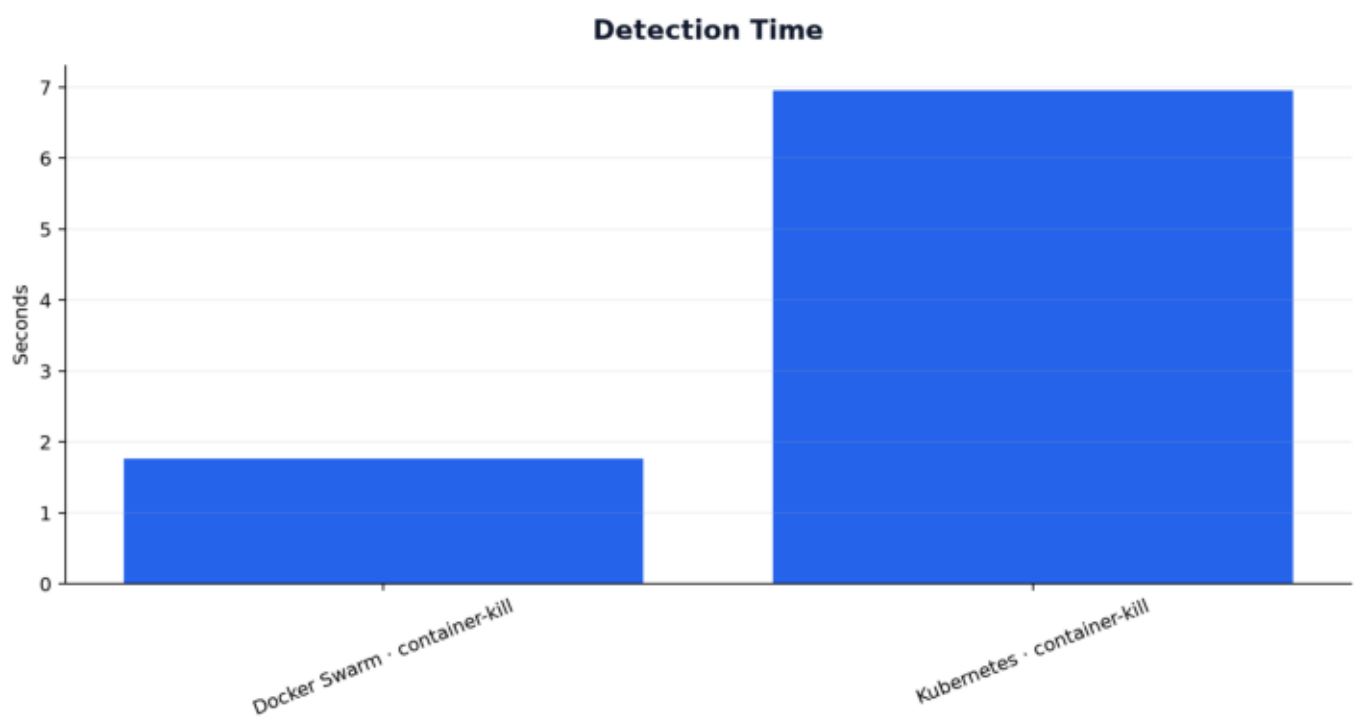
Cpu Usage

Current-session measured data



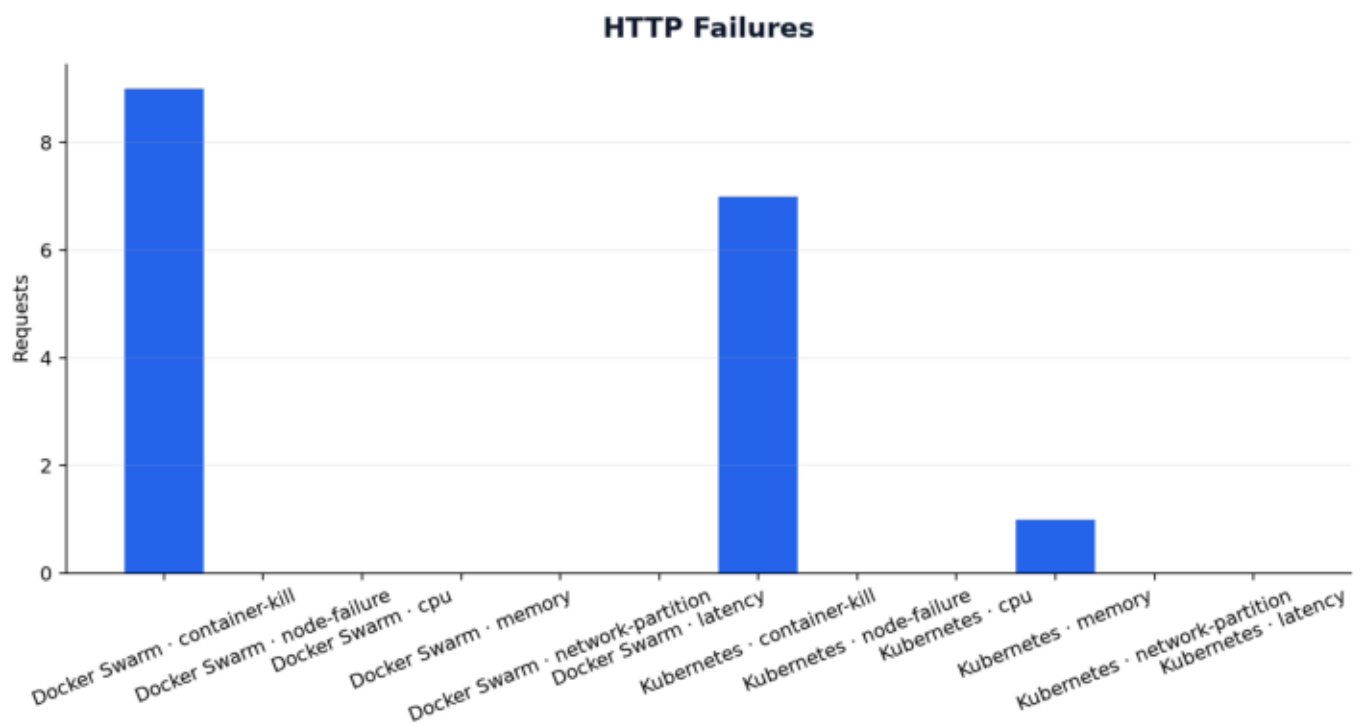
Detection Time

Current-session measured data



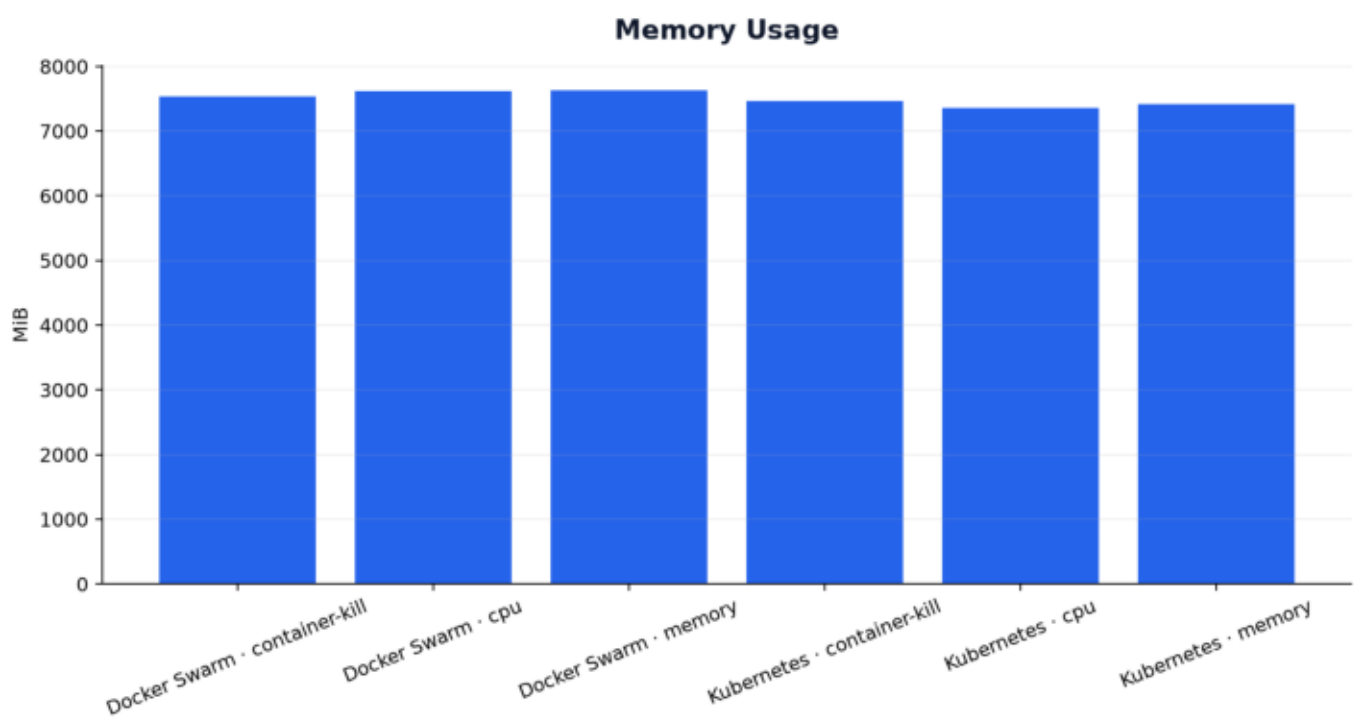
Http Failures

Current-session measured data



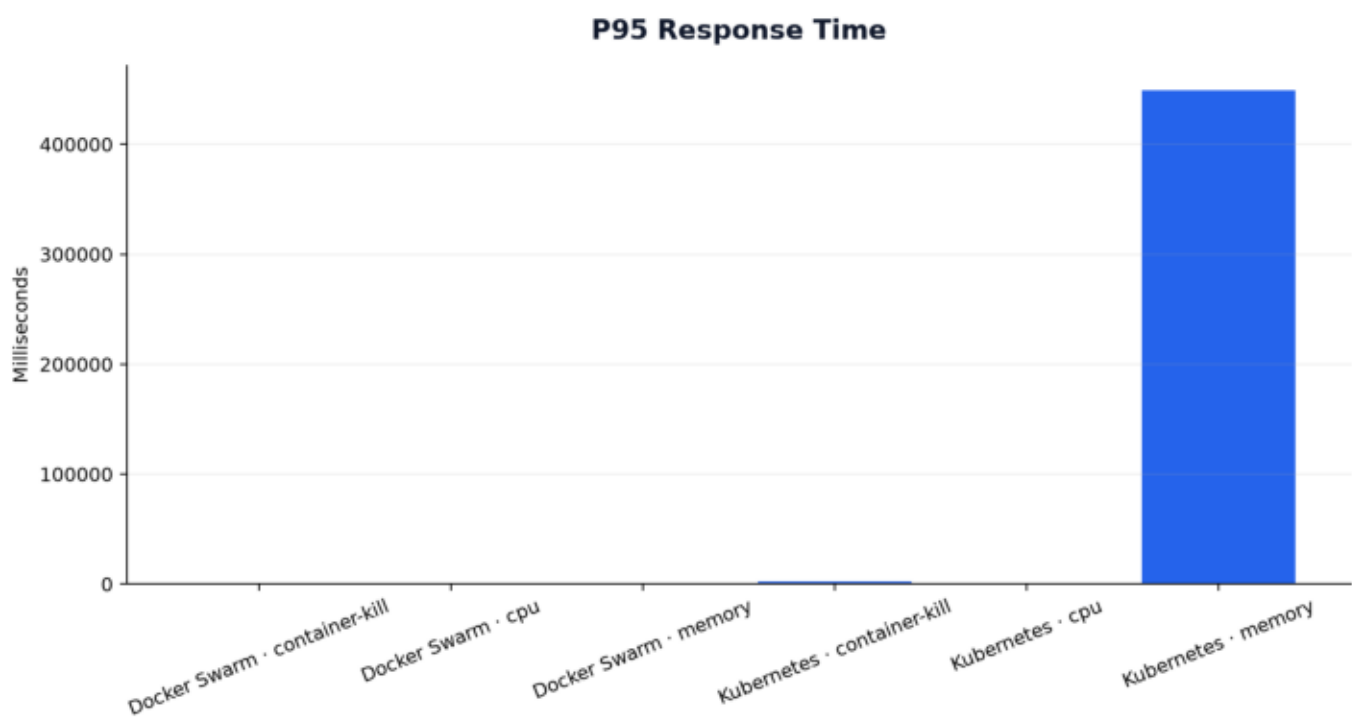
Memory Usage

Current-session measured data



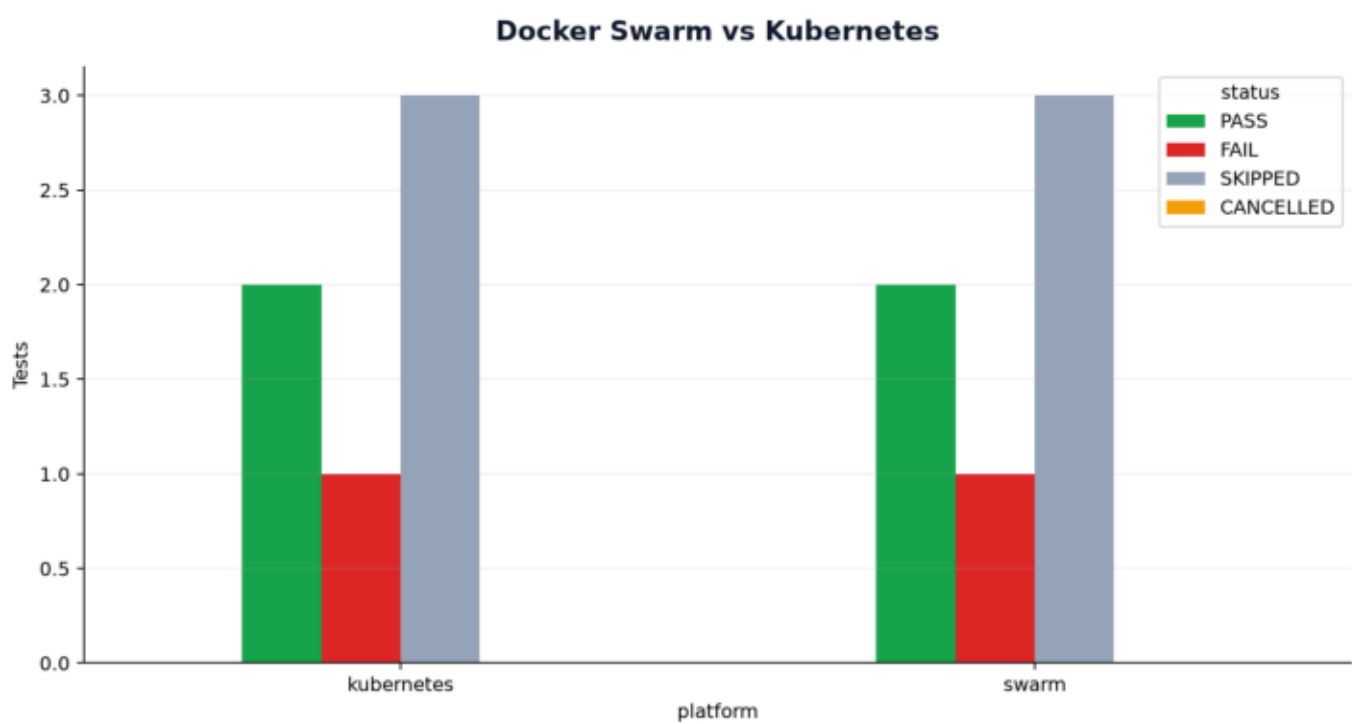
P95 Response Time

Current-session measured data



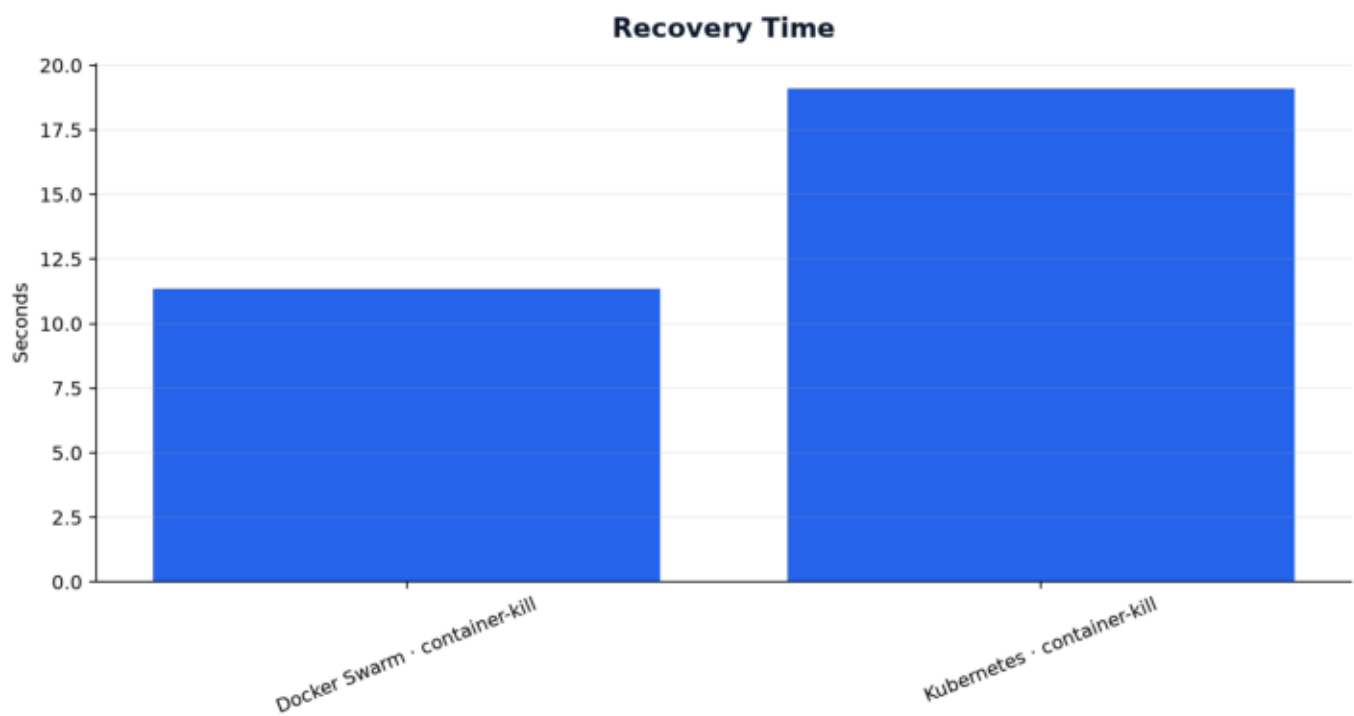
Platform Comparison

Current-session measured data



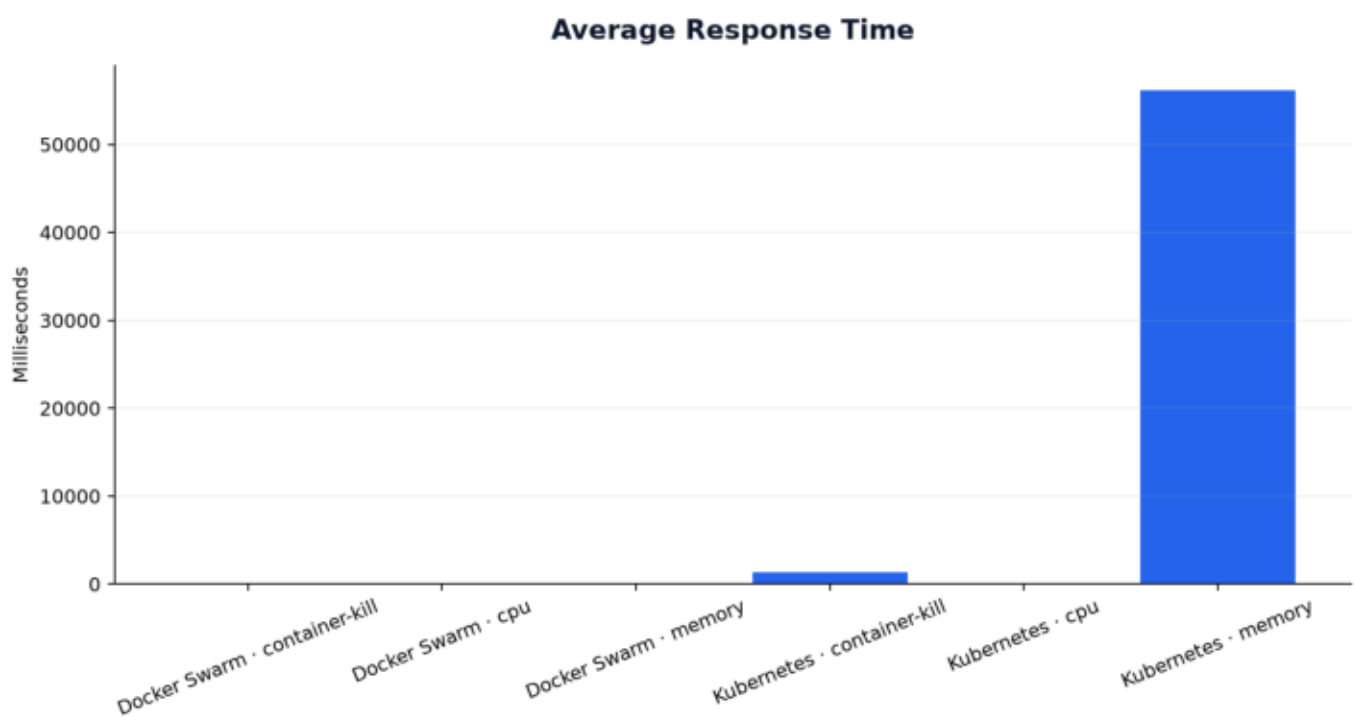
Recovery Time

Current-session measured data



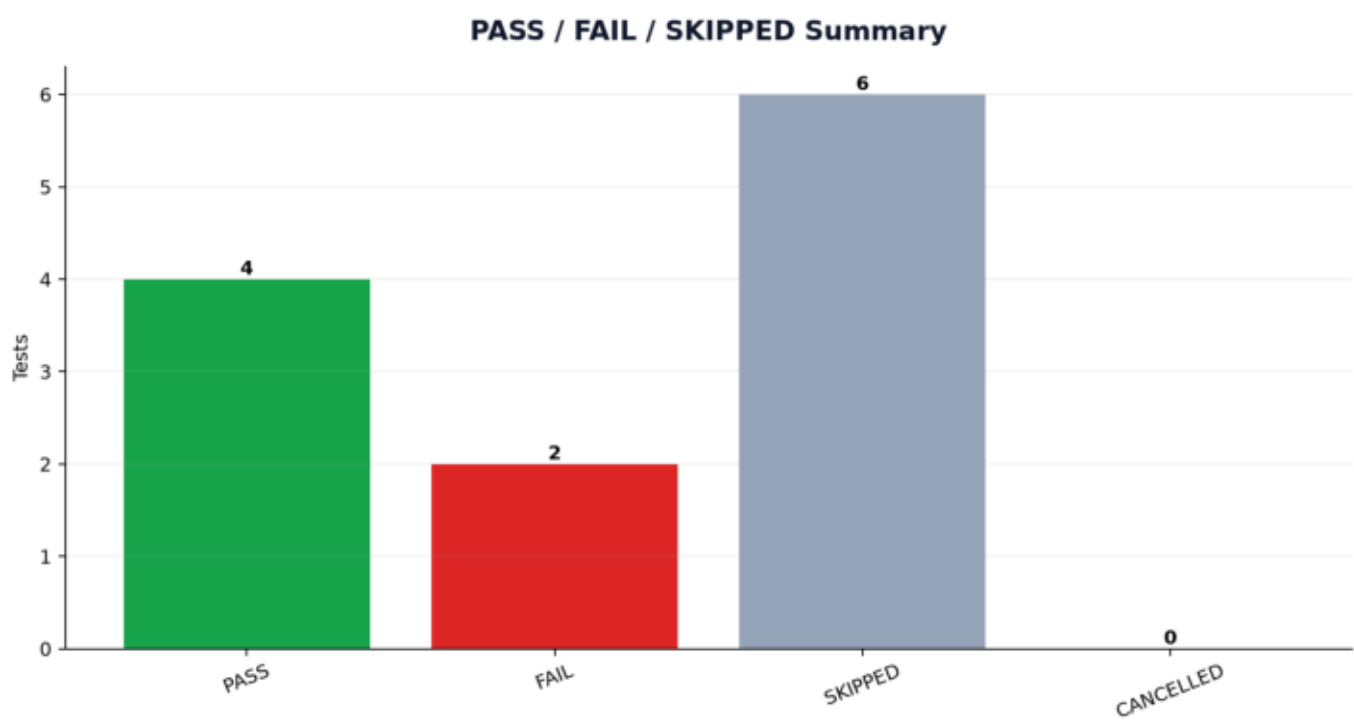
Response Time

Current-session measured data



Status Summary

Current-session measured data



Failures and Errors

Platform	Scenario	Status	Error
swarm	memory	FAIL	Swarm stress container exceeded its deadline.
kubernetes	memory	FAIL	Stress pod finished with phase Runn

Skipped Tests

Platform	Scenario	Exact reason
swarm	node-failure	Single-node Docker Swarm cannot demonstrate real node rescheduling. Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401
swarm	network-partition	CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401
swarm	latency	CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401
kubernetes	node-failure	Minikube inter-node CN/DNS does not provide Database probe to PostgreSQL authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401
kubernetes	network-partition	CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401
kubernetes	latency	CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

Swarm vs Kubernetes Comparison

Platform	FAIL	PASS	SKIPPED
kubernetes	1	2	3
swarm	1	2	3

Final Conclusion

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During this campaign, 4 of 12 scenarios passed, 2 failed and 6 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.